Chemical Bonding

(Introduction to Co-ordinate Bond)

65 HARD-level MCQs covering all 9 topics of the syllabus
With answer key & solutions

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1. Introduction

Question 1

HARD

According to the Kossel-Lewis approach, atoms combine chemically to achieve which configuration, considered particularly stable?

A. A duplet or octet configuration resembling the nearest noble gas ✓

B. A completely empty outermost shell

C. The maximum possible number of unpaired electrons

D. A configuration identical to hydrogen only

Solution: The Kossel-Lewis approach proposes that atoms combine to attain a stable electronic configuration — a duplet (2 electrons, like He) or an octet (8 electrons, like other noble gases) in their outermost shell.

Question 2

HARD

Which of the following statements about the driving force behind chemical bond formation is INCORRECT?

A. Atoms combine to attain a more stable, lower-energy state.

B. Bond formation is generally accompanied by the release of energy.

C. The tendency of atoms to combine is unrelated to their electronic configuration. ✓

D. Noble gases show little tendency to form bonds due to their already-stable configuration.

Solution: The tendency of atoms to combine is DIRECTLY related to their electronic configuration — atoms with incomplete octets are reactive, while noble gases with stable configurations are largely inert.

Question 3

HARD

Assertion (A): Ionic and covalent bonds represent two extreme, idealised types of bonding, with most real bonds showing intermediate character.
Reason (R): The electronegativity difference between the bonded atoms determines the actual ionic/covalent character of a bond.

- **A. Both A and R are true and R is the correct explanation of A. ✓**
- **B.** Both A and R are true but R is NOT the correct explanation of A.
- **C.** A is true but R is false.
- **D.** A is false but R is true.

Solution: Both statements are correct; a greater electronegativity difference pushes a bond towards more ionic character, while a smaller difference keeps it more covalent — explaining why most bonds lie between the two pure extremes.

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2. Valency

Question 4

HARD

The valency of an element in its compounds is most directly related to:

- **A.** Its atomic mass
- **B. The number of electrons in its outermost shell that participate in bonding ✓**
- **C.** Its physical state at room temperature
- **D.** Its period number in the periodic table only

Solution: Valency is determined by the number of valence (outermost shell) electrons an atom gains, loses, or shares while forming bonds.

Question 5

HARD

Which of the following correctly illustrates variable valency shown by certain elements?

A. Iron shows only +2 valency in all its compounds.

B. Sulfur shows a valency of 2 in H_2S and 6 in SF_6, depending on the compound. ✓

C. All elements show a single, fixed valency across all compounds.

D. Valency can never exceed 4 for any element.

Solution: Sulfur, like several p-block and transition elements, exhibits variable valency — 2 in H_2S but 6 in SF_6, depending on the bonding environment.

Question 6

HARD

Assertion (A): Noble gases typically show zero valency.
Reason (R): Their outermost shell already possesses a stable octet (or duplet for helium), leaving little tendency to gain, lose or share electrons.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the already-stable outer shell of noble gases is precisely why they show essentially zero valency under normal conditions.

Question 7

HARD

Which of the following correctly gives the valency of nitrogen in NH_3 and N_2O_5 respectively?

A. 3 and 5 ✓

B. 5 and 3

C. 3 and 3

D. 5 and 5

Solution: Nitrogen shows a valency of 3 in NH_3 (forming 3 N–H bonds) and a valency of 5 in N_2O_5 (its highest common oxidation state/valency in oxides).

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3. Kossel-Lewis Approach

Question 8

HARD

According to the Kossel-Lewis approach, an ionic (electrovalent) bond is formed by:

A. Sharing of an electron pair between two atoms

B. Complete transfer of one or more electrons from one atom to another ✓

C. Overlap of atomic orbitals

D. Donation of a lone pair from one atom into an empty orbital of another

Solution: An ionic bond forms via complete electron transfer from an electropositive atom (forming a cation) to an electronegative atom (forming an anion), followed by electrostatic attraction between the resulting ions.

Question 9

HARD

Which combination of factors most FAVOURS the formation of a stable ionic bond between two elements?

A. Low ionisation enthalpy of the electropositive element and high (negative) electron gain enthalpy of the electronegative element ✓

B. High ionisation enthalpy of both elements

C. Similar electronegativities of both elements

D. Both elements having nearly identical atomic radii

Solution: Ionic bond formation is favoured when one atom loses an electron easily (low ionisation enthalpy) and the other atom gains it readily with a large release of energy (high negative electron gain enthalpy).

Question 10

HARD

Lattice enthalpy of an ionic compound is defined as the energy:

A. Required to convert one mole of the solid ionic compound completely into its gaseous ions ✓

B. Released when one mole of the solid compound dissolves in water

C. Required to melt one mole of the ionic solid

D. Released when gaseous atoms combine to form the solid compound

Solution: Lattice enthalpy is the energy required to completely separate one mole of a solid ionic compound into its constituent gaseous ions, overcoming the electrostatic lattice forces.

Question 11

HARD

Which of the following compounds would be expected to show the HIGHEST lattice enthalpy, based on ionic charge and size considerations?

A. NaCl
B. MgO ✓
C. KCl
D. CsCl

Solution: Lattice enthalpy increases with higher ionic charge and smaller ionic radius. MgO, involving doubly-charged, relatively small Mg^{2}■ and O^{2}■ ions, has a much higher lattice enthalpy than the singly-charged, larger alkali halides.

Question 12

HARD

Assertion (A): Ionic compounds generally have high melting and boiling points.
Reason (R): Strong electrostatic forces of attraction exist between oppositely charged ions throughout the crystal lattice.

A. Both A and R are true and R is the correct explanation of A. ✓
B. Both A and R are true but R is NOT the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

Solution: Both statements are correct; the strong, non-directional electrostatic forces holding the entire ionic lattice together require large amounts of energy to overcome, resulting in high melting and boiling points.

Question 13

HARD

Which of the following statements about ionic compounds is INCORRECT?

A. They generally conduct electricity in the molten state or in aqueous solution.
B. They are generally soluble in polar solvents such as water.
C. They exist as discrete, individual molecules in the solid state, similar to covalent compounds. ✓
D. They exhibit a crystalline lattice structure in the solid state.

Solution: Ionic compounds do NOT exist as discrete molecules; in the solid state they form an extended three-dimensional crystal lattice of alternating cations and anions, unlike molecular covalent compounds.

Question 14

HARD

According to Fajans' rules, covalent character in a predominantly ionic bond is FAVOURED by:

- **A.** A large cation and a small anion
- **B. A small, highly-charged cation and a large anion ✓**
- **C.** Both ions being large and singly charged
- **D.** Both ions having identical charge magnitude of opposite sign only

Solution: Fajans' rules state that covalent character increases with high cationic charge, small cationic size, and large anionic size — all of which increase the polarising power of the cation and the polarisability of the anion.

Question 15

HARD

Which of the following ionic solids would be expected to show the GREATEST covalent character, according to Fajans' rules?

- **A.** NaCl
- **B.** NaF
- **C. AlCl3 ✓**
- **D.** KCl

Solution: Al^{3}■ is a small, highly-charged cation that strongly polarises the larger Cl■ anion, imparting significant covalent character to AlCl3 compared to the alkali metal halides listed.

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4. Covalent Bond

Question 16

HARD

Which of the following molecules does NOT obey the octet rule for its central atom?

- **A.** CO2
- **B. BF3 ✓**
- **C.** CH4
- **D.** NH3

Solution: Boron in BF3 has only 6 electrons around it (an electron-deficient molecule), violating the octet rule; the other three molecules have central atoms with a complete octet.

Question 17

HARD

Which of the following is an example of a molecule showing an EXPANDED octet around the central atom?

A. PCl_5 ✓

B. NH_3

C. H_2O

D. CO_2

Solution: In PCl_5, phosphorus accommodates 10 electrons around itself using its accessible d-orbitals, exceeding the normal octet of 8.

Question 18

HARD

Assertion (A): Nitric oxide (NO) is a stable odd-electron molecule.
Reason (R): It has an odd total number of valence electrons, making it impossible to satisfy the octet rule for all atoms simultaneously.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; NO has 11 valence electrons (odd number), so at least one atom must have an unpaired electron, meaning the octet rule cannot be fully satisfied — yet the molecule remains stable.

Question 19

HARD

The concept of formal charge is primarily used to:

A. Determine the most plausible/stable Lewis structure among several possible resonance structures ✓

B. Calculate the exact bond length of a molecule

C. Predict the melting point of a compound

D. Directly determine the hybridisation state of the central atom

Solution: Formal charge helps identify which of several possible Lewis structures is most stable/realistic — structures with formal charges closest to zero (and negative charges on more electronegative atoms) are generally preferred.

Question 20

HARD

What is the formal charge on the central nitrogen atom in one resonance structure of the nitrate ion (NO_3■) that has one N=O double bond and two N–O single bonds?

- **A.** 0
- **B. +1 ✓**
- **C.** –1
- **D.** +2

Solution: Formal charge = (valence electrons) – (non-bonding electrons) – ½(bonding electrons) = 5 – 0 – ½(8) = +1 for nitrogen in this resonance structure.

Question 21

HARD

Which of the following species is best described using resonance (a delocalised structure), rather than a single fixed Lewis structure?

- **A.** CH_4
- **B. O_3 (ozone) ✓**
- **C.** HCl
- **D.** NH_3

Solution: Ozone's two O–O bonds are experimentally equal in length (intermediate between single and double bond), so it cannot be represented by a single Lewis structure and is instead described as a resonance hybrid of two equivalent structures.

Question 22

HARD

Which of the following statements about resonance structures is INCORRECT?

- **A.** Resonance structures are hypothetical and do not individually exist in reality.
- **B.** The actual molecule is a hybrid with properties intermediate between the contributing structures.
- **C. Different resonance structures of a molecule can be physically isolated and studied separately. ✓**
- **D.** Resonance generally lowers the energy of a molecule, providing extra stability.

Solution: Resonance structures are theoretical constructs, not physically real or isolable species; only the resonance hybrid (the actual molecule) exists in reality.

Question 23

HARD

Which of the following diatomic species does NOT exist as a stable molecule, due to a bond order of zero?

A. He2 ✓

B. O2

C. N2

D. F2

Solution: Considering molecular orbital filling, He2 would have equal numbers of electrons in bonding and antibonding orbitals, giving a bond order of zero — so He2 does not exist as a stable molecule.

Question 24

HARD

According to the Lewis concept, a covalent bond is formed when:

A. One atom completely transfers an electron to another atom

B. Two atoms mutually share one or more electron pairs to complete their octets ✓

C. An atom loses all of its valence electrons

D. Two atoms simply come into physical contact

Solution: The Lewis concept describes a covalent bond as arising from the mutual sharing of one or more electron pairs between two atoms, allowing both to attain a stable octet (or duplet).

Question 25

HARD

As bond order increases from a single bond to a triple bond between the same pair of atoms, which trend is observed?

A. Bond order increases and bond length increases correspondingly.

B. Bond order increases while bond length decreases correspondingly. ✓

C. Bond order and bond length are unrelated to each other.

D. Bond order decreases while bond length increases.

Solution: Higher bond order (more shared electron pairs) pulls the bonded atoms closer together, so bond length DECREASES as bond order increases from single to double to triple bonds.

5. Valence Bond Theory (VBT)

Question 26

HARD

According to Valence Bond Theory, a covalent bond is formed by:

A. Complete transfer of electrons between two atoms

B. Overlap of atomic orbitals, each containing an unpaired electron of opposite spin ✓

C. Formation of a new set of hybrid orbitals only, with no orbital overlap

D. Electrostatic attraction between two complete ions

Solution: Valence Bond Theory explains covalent bond formation as the overlap of two atomic orbitals, each singly occupied by an electron of opposite spin, resulting in a localised electron pair between the nuclei.

Question 27

HARD

Which type of orbital overlap results in the formation of a pi (π) bond?

A. Head-on (axial) overlap of two s-orbitals

B. Head-on (axial) overlap of two p-orbitals along the internuclear axis

C. Sidewise (lateral) overlap of two parallel p-orbitals ✓

D. Overlap of an s-orbital with a d-orbital along the axis

Solution: A pi bond arises from the sidewise (lateral) overlap of two parallel p-orbitals, with electron density concentrated above and below (not along) the internuclear axis.

Question 28

HARD

Which of the following statements comparing sigma (σ) and pi (π) bonds is CORRECT?

A. A pi bond is generally stronger than a sigma bond due to greater orbital overlap.

B. A sigma bond involves axial overlap and is generally stronger than a pi bond, which involves lateral overlap. ✓

C. Both sigma and pi bonds involve an identical type of orbital overlap.

D. A pi bond can exist independently, without an accompanying sigma bond, between the same two atoms.

Solution: Axial (head-on) overlap in a sigma bond is more extensive/effective than the lateral overlap in a pi bond, making sigma bonds generally stronger; also, a pi bond can only form in addition to an existing sigma bond between the same atoms.

Question 29

HARD

In the carbon-carbon triple bond of acetylene (C_2H_2), the bond consists of:

A. Three sigma bonds

B. One sigma bond and two pi bonds ✓

C. Three pi bonds

D. Two sigma bonds and one pi bond

Solution: A triple bond always consists of one sigma bond (axial overlap) plus two pi bonds (lateral overlap in two mutually perpendicular planes), as seen in the C≡C bond of acetylene.

Question 30

HARD

According to Valence Bond Theory, the strength of a covalent bond formed by orbital overlap is directly related to:

A. The distance between the two nuclei, with greater distance giving stronger bonds

B. The extent (degree) of orbital overlap between the combining atomic orbitals ✓

C. The total number of electrons in the atom, regardless of overlap

D. The atomic mass of the combining atoms

Solution: Greater orbital overlap results in greater electron density between the nuclei and hence a stronger, more stable covalent bond.

Question 31

HARD

Which of the following statements about orbital overlap in Valence Bond Theory is INCORRECT?

A. Greater overlap between orbitals leads to a stronger bond.

B. Only atomic orbitals containing unpaired electrons can participate in overlap to form a covalent bond.

C. The electron density in a pi bond is symmetrical about the internuclear axis, exactly as in a sigma bond. ✓

D. Overlapping orbitals must have appropriate symmetry and comparable energies for effective bond formation.

Solution: In a sigma bond, electron density IS symmetrical about the internuclear axis; however, in a pi bond the electron density lies above and below the axis (not symmetric about it), making this statement incorrect.

Question 32

HARD

Assertion (A): A sigma bond permits free rotation of the bonded atoms around the bond axis.
Reason (R): The electron cloud of a sigma bond is symmetrical about the internuclear axis, so rotation does not disrupt orbital overlap.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; because the sigma bond's electron density is cylindrically symmetric about the bond axis, rotating the atoms around this axis does not reduce orbital overlap, permitting free rotation.

Question 33

HARD

Which of the following statements about pi bonds and molecular rigidity is CORRECT?

A. A pi bond permits completely free rotation around the bond axis, just like a sigma bond.

B. A pi bond restricts free rotation, since rotation would disrupt the lateral overlap of the parallel p-orbitals. ✓

C. Pi bonds have no influence on molecular geometry or rigidity.

D. Pi bonds can only form between atoms of the same element.

Solution: Rotating around a bond containing a pi component would misalign the parallel p-orbitals, breaking the lateral overlap — hence pi bonds restrict free rotation, giving rise to phenomena like cis-trans isomerism.

6. Hybridisation

Question 34

HARD

Which type of hybridisation is exhibited by the carbon atom in methane (CH_4)?

A. sp

B. sp^2

C. sp^3 ✓

D. sp^3d

Solution: Carbon in CH_4 mixes one s and three p orbitals to form four equivalent sp^3 hybrid orbitals, directed towards the corners of a tetrahedron.

Question 35

HARD

Which hybridisation and resulting molecular shape is shown by $BeCl_2$?

A. sp hybridisation, linear shape ✓

B. sp2 hybridisation, trigonal planar shape

C. sp3 hybridisation, tetrahedral shape

D. sp3d hybridisation, trigonal bipyramidal shape

Solution: Beryllium in $BeCl_2$ undergoes sp hybridisation (mixing one s and one p orbital), giving two hybrid orbitals oriented at 180°, resulting in a linear molecular shape.

Question 36

HARD

The hybridisation of the central phosphorus atom in PCl_5 is:

A. sp3

B. sp3d ✓

C. sp3d2

D. sp3d3

Solution: Phosphorus in PCl_5 undergoes sp3d hybridisation, forming five sigma bonds arranged in a trigonal bipyramidal geometry.

Question 37

HARD

The hybridisation of sulfur in SF_6 is:

A. sp3

B. sp3d

C. sp3d2 ✓

D. sp3d3

Solution: Sulfur in SF_6 undergoes sp3d2 hybridisation, forming six equivalent sigma bonds arranged octahedrally.

Question 38

HARD

Which of the following correctly matches a hybridisation state with its ideal bond angle (assuming no lone pair distortion)?

A. sp – 180° ✓

B. sp2 – 90°

C. sp3 – 120°

D. sp3d2 – 60° only

Solution: sp hybridisation gives a linear arrangement with a 180° bond angle. (sp2 actually gives 120°, sp3 gives 109.5°, and sp3d2 gives 90°/180° — so only option A is correctly matched.)

Question 39

HARD

Which of the following molecules involves sp2 hybridisation of its central atom?

A. CH_4

B. BF_3 ✓

C. $BeCl_2$

D. PCl_5

Solution: Boron in BF_3 mixes one s and two p orbitals to form three sp2 hybrid orbitals, giving a trigonal planar geometry with 120° bond angles.

Question 40

HARD

Assertion (A): In NH_3, the observed H-N-H bond angle (~107°) is slightly LESS than the ideal tetrahedral angle (109.5°), even though nitrogen is sp3 hybridised.
Reason (R): The lone pair on nitrogen exerts greater repulsion on the bonding pairs than bond pairs exert on one another, compressing the bond angle.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the stronger lone pair–bond pair repulsion (compared to bond pair–bond pair repulsion) pushes the N–H bonds slightly closer together, explaining the reduced bond angle.

Question 41

HARD

Which of the following statements about hybridisation is INCORRECT?

- **A.** Hybrid orbitals form by mixing atomic orbitals of comparable energy on the same atom.
- **B.** The number of hybrid orbitals formed equals the number of atomic orbitals that mix together.
- **C. Hybrid orbitals give WEAKER bonds compared to pure, unhybridised atomic orbitals. ✓**
- **D.** Hybrid orbitals are oriented in specific directions to minimise electron pair repulsion.

Solution: Hybrid orbitals actually form STRONGER bonds than pure atomic orbitals, owing to their greater directional character and more effective overlap — making this statement incorrect.

Question 42

HARD

The hybridisation state of each carbon atom in a molecule containing a C≡C triple bond (e.g., acetylene) is:

- **A. sp ✓**
- **B.** sp2
- **C.** sp3
- **D.** sp3d

Solution: Each carbon in a C≡C triple bond is sp hybridised (one sigma bond via sp-sp overlap, plus two pi bonds from unhybridised p-orbitals).

Question 43

HARD

Which shape results from sp3d hybridisation when all five hybrid orbitals form bond pairs with no lone pairs present?

- **A.** Tetrahedral
- **B. Trigonal bipyramidal ✓**
- **C.** Octahedral
- **D.** Square pyramidal

Solution: sp3d hybridisation with 5 bond pairs (as in PCl_5) gives a trigonal bipyramidal molecular geometry.

Question 44

HARD

Which of the following molecules shows sp^3 hybridisation of the central atom, but a BENT (angular) rather than tetrahedral molecular shape, due to lone pairs?

A. CH_4

B. H_2O ✓

C. $BeCl_2$

D. BF_3

Solution: Oxygen in H_2O is sp^3 hybridised with 2 bond pairs and 2 lone pairs; since molecular shape considers only the positions of bonded atoms, the two lone pairs give water a bent shape rather than a tetrahedral one.

Question 45

HARD

Assertion (A): sp^3d^2 hybridisation results in an octahedral molecular geometry when all six hybrid orbitals form bond pairs.
Reason (R): sp^3d^2 hybridisation involves the mixing of one s, three p, and two d orbitals, giving six equivalent hybrid orbitals directed towards the corners of an octahedron.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; the geometric arrangement of the six sp^3d^2 hybrid orbitals directly gives rise to the octahedral shape when all are occupied by bond pairs (as in SF_6).

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7. VSEPR Theory

Question 46

HARD

According to VSEPR theory, the primary factor determining the geometry of a molecule is:

A. The atomic masses of the bonded atoms

B. Minimisation of repulsion between electron pairs (bonding and lone) around the central atom ✓

C. The type of hybridisation exclusively, independent of electron pair repulsion

D. The polarity of the individual bonds

Solution: VSEPR (Valence Shell Electron Pair Repulsion) theory is based on the principle that electron pairs around a central atom arrange themselves in space to minimise mutual repulsion, determining the resulting molecular geometry.

Question 47

HARD

According to VSEPR theory, the correct order of decreasing repulsive strength between electron pairs is:

A. Bond pair-bond pair > Lone pair-bond pair > Lone pair-lone pair

B. Lone pair-lone pair > Lone pair-bond pair > Bond pair-bond pair ✓

C. Lone pair-bond pair > Lone pair-lone pair > Bond pair-bond pair

D. All three types of repulsion are of equal magnitude

Solution: Lone pairs, being more diffuse and closer to the nucleus, exert the strongest repulsion, giving the order: lone pair-lone pair > lone pair-bond pair > bond pair-bond pair.

Question 48

HARD

The molecular shape of NH_3, according to VSEPR theory, is:

A. Trigonal planar

B. Pyramidal (trigonal pyramidal) ✓

C. Tetrahedral

D. Linear

Solution: Nitrogen in NH_3 has 3 bond pairs and 1 lone pair; considering only the positions of bonded atoms, the resulting molecular shape is trigonal pyramidal, not tetrahedral.

Question 49

HARD

Which of the following molecules has a SQUARE PLANAR shape, according to VSEPR theory?

A. CH_4

B. XeF_4 ✓

C. NH_3

D. PCl_5

Solution: XeF_4 has 4 bond pairs and 2 lone pairs around the central Xe atom (sp^3d^2 hybridised); the two lone pairs occupy axial positions, leaving the four F atoms in a square planar arrangement.

Question 50

HARD

Which pair of molecules share the SAME basic molecular shape, despite differing central atoms?

A. $BeCl_2$ and CO_2 (both linear) ✓

B. NH_3 and CH_4 (both tetrahedral)

C. BF_3 and NH_3 (both pyramidal)

D. H_2O and CH_4 (both bent)

Solution: Both $BeCl_2$ (sp hybridised, no lone pairs) and CO_2 (sp hybridised, no lone pairs) adopt a linear molecular shape.

Question 51

HARD

The shape of the SF_4 molecule (sp^3d hybridised, with one lone pair) is best described as:

A. Trigonal bipyramidal

B. See-saw (irregular tetrahedron) ✓

C. Square planar

D. Trigonal pyramidal

Solution: SF_4 has 4 bond pairs and 1 lone pair; the lone pair occupies an equatorial position of the trigonal bipyramidal electron geometry, distorting the molecular shape into a see-saw arrangement.

Question 52

HARD

Which of the following correctly explains why the H-O-H bond angle in water (104.5°) is smaller than the H-N-H bond angle in ammonia (107°)?

A. Oxygen has two lone pairs (vs. one in nitrogen), causing greater lone pair-bond pair repulsion and further bond angle compression. ✓

B. Oxygen is a larger atom than nitrogen.

C. Water has a higher molecular mass than ammonia.

D. The O-H bond is longer than the N-H bond.

Solution: With two lone pairs (compared to nitrogen's one), oxygen exerts stronger cumulative lone pair-bond pair repulsion, compressing the H-O-H angle further below the ideal tetrahedral value than in ammonia.

Question 53

HARD

Which of the following statements about VSEPR theory is INCORRECT?

A. Multiple bonds (double/triple) are treated as a single electron-dense region for geometry purposes.

B. Lone pairs occupy more space than bonding pairs, exerting greater repulsion.

C. The theory accurately predicts bond lengths but not molecular shape. ✓

D. Electron pairs around a central atom arrange to minimise mutual repulsion.

Solution: VSEPR theory is specifically used to predict molecular SHAPE based on electron pair repulsion; it does not directly predict bond lengths, making this statement incorrect.

Question 54

HARD

The shape of ClF3 (3 bond pairs, 2 lone pairs around the central Cl, sp3d hybridised) is:

A. T-shaped ✓

B. Trigonal planar

C. See-saw

D. Pyramidal

Solution: With both lone pairs occupying equatorial positions of the trigonal bipyramidal electron geometry, the three fluorine atoms are left in a T-shaped molecular arrangement.

Question 55

HARD

Which of the following molecules is CORRECTLY matched with its molecular shape according to VSEPR theory?

- **A.** PCl_5 – Octahedral
- **B.** SF_6 – Trigonal bipyramidal
- **C. XeF_2 – Linear ✓**
- **D.** BF_3 – Pyramidal

Solution: XeF_2 has 2 bond pairs and 3 lone pairs (sp^3d hybridised); with all three lone pairs equatorial, the molecule is linear. (PCl_5 is actually trigonal bipyramidal, SF_6 is octahedral, and BF_3 is trigonal planar.)

8. Bond Parameters

Question 56

HARD

Bond length is defined as:

- **A. The equilibrium distance between the nuclei of two covalently bonded atoms ✓**
- **B.** The energy required to break a chemical bond
- **C.** The angle between two adjacent bonds at a common atom
- **D.** The total number of electrons shared between two atoms

Solution: Bond length is the average equilibrium distance between the nuclei of two atoms joined by a covalent bond.

Question 57

HARD

Which of the following correctly relates bond order to bond length and bond enthalpy?

- **A.** As bond order increases, bond length increases and bond enthalpy decreases.
- **B. As bond order increases, bond length decreases and bond enthalpy increases. ✓**
- **C.** Bond order has no relationship with either bond length or bond enthalpy.
- **D.** As bond order increases, both bond length and bond enthalpy increase.

Solution: Higher bond order means more shared electron pairs pulling the nuclei closer (shorter bond length) and requiring more energy to break (higher bond enthalpy).

Question 58

HARD

Dipole moment is a vector quantity; by convention, its direction is represented:

A. From the negative end to the positive end of the dipole

B. From the positive end to the negative end of the dipole ✓

C. Perpendicular to the bond axis

D. Dipole moment has no defined direction, only magnitude

Solution: By convention (in chemistry), the dipole moment vector points from the positive end of the dipole towards the negative end.

Question 59

HARD

Which of the following molecules has a ZERO net dipole moment, despite having individually polar bonds, due to its symmetrical geometry?

A. CO_2 ✓

B. H_2O

C. NH_3

D. HCl

Solution: Although each C=O bond in CO_2 is polar, the molecule's linear, symmetrical geometry causes the two bond dipoles to exactly cancel, giving a net dipole moment of zero.

Question 60

HARD

The percentage ionic character of a covalent bond can be estimated using the ratio of:

A. The observed dipole moment to the dipole moment calculated assuming 100% ionic character ✓

B. Bond length to bond energy

C. Atomic mass to atomic number

D. Ionisation enthalpy to electron gain enthalpy

Solution: Percentage ionic character is estimated by comparing the experimentally observed dipole moment with the theoretical dipole moment expected for a fully (100%) ionic bond of the same bond length.

Question 61

HARD

Assertion (A): The H-F bond has a higher percentage ionic character than the H-I bond.
Reason (R): The electronegativity difference between H and F is greater than that between H and I.

A. Both A and R are true and R is the correct explanation of A. ✓

B. Both A and R are true but R is NOT the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Solution: Both statements are correct; a larger electronegativity difference between bonded atoms directly correlates with greater ionic character, explaining why H-F is more ionic than H-I.

Question 62

HARD

Which of the following does NOT directly influence bond enthalpy (bond dissociation energy)?

A. Bond length (shorter bonds are generally stronger)

B. Bond order (higher order generally means stronger bonds)

C. Size of the bonded atoms (smaller atoms generally form stronger bonds)

D. The colour of the compound formed ✓

Solution: Bond enthalpy depends on bond length, bond order and atomic size (which affect orbital overlap), but has no direct relationship with the colour of the resulting compound.

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9. Co-ordinate (Dative) Bond

Question 63

HARD

A coordinate (dative) covalent bond is formed when:

A. Both bonded atoms contribute one electron each to the shared pair

B. One atom (donor) contributes both electrons of the shared pair, while the other atom (acceptor) contributes none ✓

C. Electrons are completely transferred from one atom to another, forming ions

D. Two atoms of identical electronegativity share electrons equally

Solution: In a coordinate bond, one atom (the donor) supplies both electrons of the shared pair from its lone pair, while the acceptor atom provides an empty orbital to receive them.

Question 64

HARD

In the formation of the ammonium ion (NH4■) from NH3 and H■, the coordinate bond is formed by:

- **A. Nitrogen donating its lone pair of electrons into the empty orbital of H■ ✓**
- **B.** Hydrogen donating an electron to nitrogen
- **C.** Equal sharing of a newly formed electron pair contributed by both nitrogen and the hydrogen ion
- **D.** Complete transfer of an electron from H■ to nitrogen

Solution: Nitrogen's lone pair (from NH3) is donated into the empty 1s orbital of H■, forming the fourth N-H bond in NH4■ as a coordinate bond.

Question 65

HARD

Which of the following statements about coordinate bonds is CORRECT?

- **A.** Once formed, a coordinate bond remains chemically and physically distinguishable from a normal covalent bond within the same molecule/ion.
- **B. A coordinate bond requires one atom to have a lone pair and the other to have a vacant orbital capable of accepting that pair. ✓**
- **C.** Coordinate bonds cannot form between a neutral molecule and an ion.
- **D.** Coordinate bonds are always weaker than ionic bonds.

Solution: Once formed, all four N-H bonds in NH4■ are experimentally indistinguishable (identical bond length and strength), so option A is incorrect. The essential requirement for coordinate bond formation — a lone pair donor and a vacant-orbital acceptor — is correctly described in option B.
